Solar PV System Specification

Material Costing & Return Analysis | Standard 3-Bedroom Home (RSA)

TARGET SYSTEM
5kW Inverter + 10kWh Battery

ESTIMATED TURNKEY COST
R 118,500
Includes VAT, Scaffolding & CoC

EST. MONTHLY SAVINGS
R 2,450
Based on Eskom Block 2 rates

SIMPLE PAYBACK
4.0 Years
Assuming 12.5% annual tariff hikes

DAILY YIELD
22–26 kWh
Average sun hours in Gauteng/KZN

1. BILL OF MATERIALS & TURNKEY COST BREAKDOWN

ITEM DESCRIPTION	BRAND / SPEC	QTY	UNIT PRICE	TOTAL (ZAR)
GENERATION & STORAGE CORE COMPONENTS				
Sunsynk 5kW Hybrid Inverter (Single Phase)	Sunsynk-5.5-K-SG01LP1	1	R 22,500	R 22,500
5.12kWh LiFePO4 Lithium Battery 48V	Hubble AM-2 / Sunsynk / Freedom Won	2	R 19,500	R 39,000
550W Tier-1 Monocrystalline Solar Panels	JA Solar / Canadian Solar / Jinko	8	R 1,850	R 14,800
BALANCE OF SYSTEM, BALANCE OF PLANT & PROTECTION				
Aluminium Roof Rail Mounting Kit (Tile/IBR Roof)	Renusol / K2 Systems Structuring	1	R 4,800	R 4,800
AC Protection Board (Changeover Switch, Surge Protection)	Pre-wired 12-way IP65 DB Enclosure	1	R 3,500	R 3,500
DC Combiner & Disconnect (Fused Isolators, Type 2 SPD)	600V Solar DC Protection Box	1	R 2,800	R 2,800
Solar Cable (6mm² UV Double Insulated red/black)	4mm²/6mm² Solar PV Flex Cable (50m)	1	R 1,850	R 1,850
Battery Cable Set (35mm² Flexible Copper & Lugs)	Heavy duty 200A battery interconnects	1	R 1,450	R 1,450
Conduit, Trunking, Glands & Earthing Rod Kit	25mm PVC Conduit, Earth Wire & Spike	1	R 2,300	R 2,300
PROFESSIONAL SERVICES, LABOUR & COMPLIANCE				
Qualified PV Installation Labour & Engineering	Registered SAPVIA/PV GreenCard Team	1	R 18,000	R 18,000
Electrical Certificate of Compliance (CoC) & Sign-off	Registered Master Electrician Inspection	1	R 3,500	R 3,500
Municipal SSEG Registration & Application Fee	City Power / Cape Town / Tshwane SSEG	1	R 4,000	R 4,000
TOTAL ESTIMATED INVESTMENT (INCL. VAT)				**R 118,500**

Cost Disclaimer: Prices are estimates based on standard RSA market rates (2025/2026). Costs vary based on roof pitch, distance from DB to solar array, and regional municipal application fees. Geyser non-essential load integration is assumed.

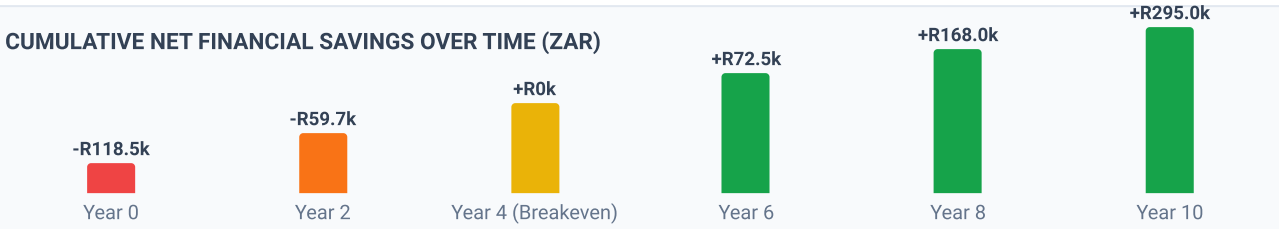
2. SYSTEM SIZING & FINANCIAL RETURN ANALYSIS

Estimated Energy Load Coverage

- **Daytime Coverage (100% Solar):** Refrigerator, home office, TV/entertainment, Wi-Fi, lights, washing machine, and dishwasher.
- **Nighttime Coverage (Battery 10kWh):** 80% Depth of Discharge provides 8kWh usable power. Covers ambient lighting, refrigeration, security systems, and ceiling fans overnight.
- **Non-Essential Exclusions:** Electric geyser, stove/oven, and air conditioners remain on utility or controlled via smart relay during peak battery use.

System Energy Yield Metrics

- **Installed PV Capacity:** 4.40 kWp (8 × 550W panels)
- **Avg. Peak Sun Hours:** 5.2 hours/day (Gauteng baseline)
- **Est. Daily Generation:** 22.8 kWh / day
- **Est. Annual Generation:** ~8,300 kWh
- **Offset of Municipal Bill:** ~75%–85% reduction in total monthly electricity units consumed.



3. KEY FINANCIAL & TECHNICAL RECOMMENDATIONS

Critical Technical Considerations

- **Geyser Optimization:** Install a Geyserwise controller or smart timer (approx. R2,500 extra) to shift geyser heating to peak solar production hours (11:00–14:00).
- **DB Board Separation:** Ensure your electrician splits essential loads (lights, plugs, internet) from non-essential loads (oven, pool pump, geyser).
- **Roof Inspection:** Ensure roof trusses are structurally sound prior to mounting panel rails.

Regulatory & Warranty Guidelines

- **Inverter Warranty:** 5 Years standard (extendable to 10 years upon registration).
- **Battery Warranty:** 10 Years / 6,000 cycles at 80% Depth of Discharge.
- **Solar Panel Performance:** 12 Year product warranty, 25-30 Year linear power output guarantee.
- **Compliance:** Ensure your installer provides an Electrical CoC and files the SSEG application to avoid municipal fines or disconnects.